An End-to-End Interpretable Machine Learning Pipeline for Maternal Health Risk Prediction

***Abstract*—Preventable maternal mortality remains a major global issue, emphasizing the need for reliable and straightforward early-warning systems. Maternal mortality continues to be a critical health challenge worldwide, with progress in reducing it having plateaued. Machine learning (ML) models hold considerable promise for early risk stratification; however, their "black box" nature has hindered their adoption by doctors. This study fills a significant gap by creating and testing a robust, transparent, and high-performing pipeline for determining the likelihood of a mother experiencing health problems (Low, Mid, High). We highlight three main innovations: a systematic feature engineering process that generates new interaction features; a comprehensive comparison of different classifiers using 10-fold stratified cross-validation; and the adoption of Explainable AI (XAI) tools, such as SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME), to ensure the final model is fully interpretable. On a publicly available dataset of 1014 patient records, our best model, a Random Forest classifier, achieved an overall accuracy and Macro F1-Score of 86% on an unseen test set. The model demonstrated good clinical utility with a recall of 96% for the high-risk class, thus decreasing the chance of missing cases.Importantly, XAI analysis revealed that a newly engineered feature representing the interaction between age and blood sugar (Age_BS_Interaction) was the most influential predictor. This work presents a comprehensive framework for developing prediction models that are both accurate and trustworthy, with practical clinical utility, thereby addressing a significant obstacle to the successful application of AI in maternal healthcare.**

***Index Terms*—Maternal health, machine learning, explainable AI, risk prediction, Categorical Boosting (CatBoost), SMOTEENN, clinical decision support.**

I. Introduction

MATERNAL mortality remains a major global health challenge. Although the worldwide Maternal Mortality Ratio (MMR) dropped significantly from 2000 to 2023, this progress has alarmingly stalled, highlighting an urgent need for new strategies to improve maternal health [1]. The problem is even more visible in the developed countries. For example, the maternal death rate in the United States reached 13.56 deaths per 100,000 births between 2020 and 2022, a level not seen in two decades [2]. In 2023, the US MMR was 18.6 deaths per 100 000 live births, with a much higher rate for non-Hispanic Black women (50.3) than for non-Hispanic White (14.5) or Hispanic (12.4) women [3].

These averages mask wide variation; women from black ethnic minorities are nearly three times more likely to die than white women, and women from the most deprived areas have double the mortality of those from the least deprived areas [2]. This ongoing inequality points to large systemic problems in the delivery of equitable healthcare [4]. These challenges have resulted in an increasing use of machine learning (ML) by researchers to identify complex, non-linear patterns in health data that are missed by traditional statistical methods.

We used the same public data set and several studies have shown success in using ML to predict maternal health risks. For instance, Pavagada and Vemuri got 84.12% accuracy using an Ensemble Bagged Trees model [5]. Similarly, Pi et al. compared six algorithms and reported that Multilayer Perceptron (MLP) performed best with an accuracy of 82% [6]. Also, Adeola et al. achieved an accuracy of 84% using Extreme Gradient Boosting (XGBoost) and Stacking Classifier, and demonstrated the practical usefulness by building a web application that used the XGBoost model for deployment [7]. Table I summarises the methodologies, performance indicators, and limitations of the reviewed literature.

But a survey of existing research reveals some major gaps. One big problem is that there is no systematic search for feature engineering, and standard feature sets are relied on, without producing new variables that capture complex physiological interactions.

More importantly, the model suffers from a serious interpretability problem. Previous research often results in "black box" models where the decision-making processes are not clear, limiting their clinical utility as healthcare providers are not able to verify or trust these outcomes. Moreover, several studies evaluate models on only one train-test split, which may not be a good indicator of their generalisation capacity. The main contributions of this paper are as follows.

- Developed a prediction pipeline using systematic feature engineering and sophisticated data balancing strategies to address problems with class imbalance.
- We utilised an extensive 10-fold cross-validation on various machine learning algorithms, including state-of-the-art ensemble techniques, to select the most effective model with real-world data.
- Applied Explainable AI (XAI) methods such as SHAP and LIME to provide complete transparency of the model, confirming the accuracy and reliability of predictions for clinical decision making.

Section II describes our materials and methods. Section III discusses the empirical results. Section IV interprets the findings, and Section V delivers the conclusion.

II. Materials & Methods

A. Research Goal and Methodological Framework

The main goal of this research is to create and evaluate a robust machine learning classification model to predict health

TABLE I: Literature Review Summary

Existing Work	Method	Performance	Limitation	Future Work
Pavagada & Vemuri [5]	Ensemble Bagged Trees, k-NN, Neural Networks	84.12% accuracy with Bagged Trees	Standard feature set; limited model comparison	Explore more complex models
Pi et al. [6]	Six ML algorithms with SMOTE	82% accuracy with MLP	Small dataset; no feature engineering	Clinical integration and validation
Adeola et al. [7]	Random Forest and other models; developed a web app	84% accuracy with XGBoost & Stacking Classifier	Focus on deployment; less emphasis on interpretability	Enhance model transparency and trust

risk level in pregnant patients (Low, Mid, or High Risk). The method involves a detailed, multi-step process, as shown in Fig. 1, that ensures precise data handling, careful model choice, extensive performance evaluation, and straightforward interpretation of the predictions.

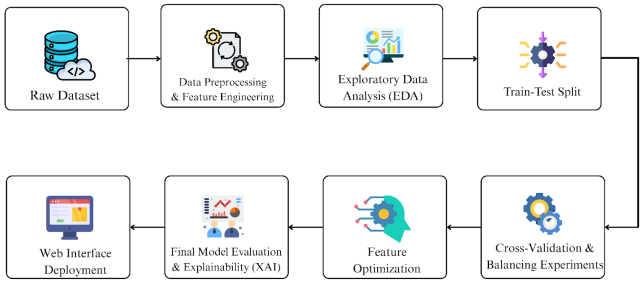


Fig. 1: Overview of the end-to-end methodological pipeline.

B. Data Source and Preprocessing

The "Maternal Health Risk Data Set," a publicly accessible dataset from the UCI Machine Learning Repository, is used in this study [8]. The original dataset included 1,014 cases and consisted of seven features representing important clinical indicators: Age (10–70 years), Systolic Blood Pressure (70–160 mmHg), Diastolic Blood Pressure (49–100 mmHg), Blood Sugar (BS, 6.0–19.0 mmol/L), Body Temperature (98.0–103.0 °F), HeartRate (60–90 bpm) and RiskLevel. A description of these features is available in Table II.

TABLE II: Dataset Feature Description

Feature Name	Type	Description	Value Range
Age	Numeric	Age of the patient in years	10–70
SystolicBP	Numeric	Upper value of Blood Pressure (mmHg)	70–160
DiastolicBP	Numeric	Lower value of Blood Pressure (mmHg)	49–100
BS	Numeric	Blood Sugar level (mmol/L)	6.0–19.0
BodyTemp	Numeric	Body Temperature in Fahrenheit	98.0–103.0
HeartRate	Numeric	Resting heart rate (beats per minute)	60–90
RiskLevel	Categorical	Final risk diagnosis	low, mid, high risk

The target variable, RiskLevel, has three categories: "low risk," "mid risk," and "high risk." The data-cleaning process included detecting and replacing biologically implausible artifacts, such as a HeartRate of 7, with the column median. The categorical target variable was encoded numerically as low risk (0), mid risk (1), and high risk (2).

C. Feature Engineering and Data Splitting

Three new characteristics were introduced to better capture complex physiological relationships:

1) PulsePressure = SystolicBP – DiastolicBP
2) Age_BS_Interaction = Age × BS
3) Systolic_Age_Interaction = SystolicBP × Age

A stratified split was used to maintain the class distribution across both subsets of the entire dataset, which now included these engineered features, into an 80% training set and a 20% hold-out testing set.

D. Exploratory Data Analysis

We explored feature distributions and their relationship with the target variable using exploratory data analysis (EDA). Fig. 2 shows box plots of feature distributions at different risk levels. Engineered features, especially Age_BS_Interaction and PulsePressure, clearly separate between risk groups [9]. A correlation heatmap was generated in Fig. 3 to measure linear dependencies, and it validated the strong correlation between the newly engineered features and the RiskLevel target [10].

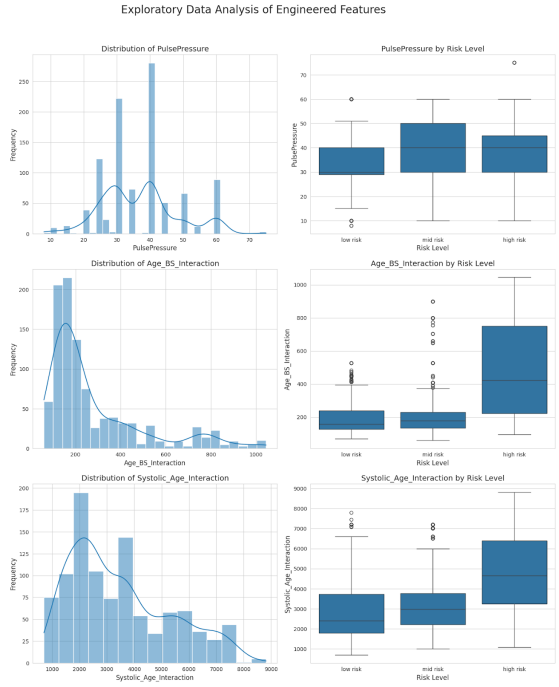


Fig. 2: Box plots of feature distributions across risk levels.

E. Model Construction and Selection

The whole experimental setup was designed to find the best classification model. We experimented with a diverse set of algorithms such as Logistic Regression [11], Support Vector Machines (SVM) [12] and more advanced ensemble techniques such as Random Forest [13], Light Gradient Boosting Machine (LightGBM) [14], Bagging [15], Voting [16] and Stacking [17]. To deal with the natural class imbalance in the training data, several methods were systematically tested: no balancing (baseline), SMOTE (over-sampling) [18], and

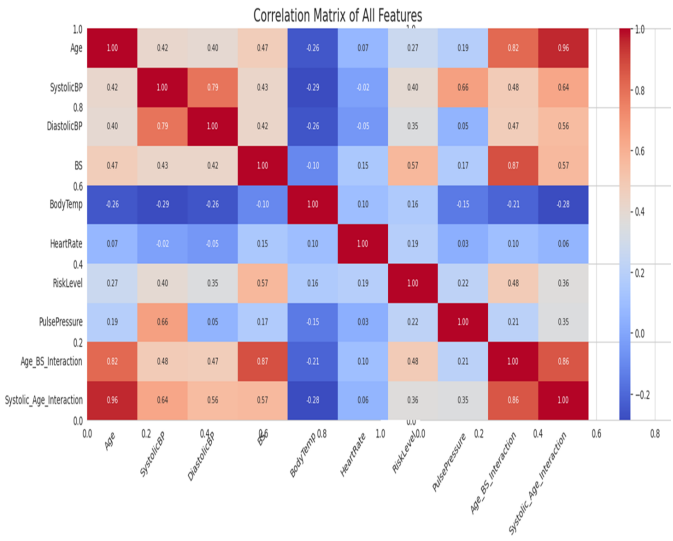


Fig. 3: Correlation heatmap of dataset features and the RiskLevel target.

adding class weights directly to the classifier [19]. We used 10-fold stratified cross-validation [20] to carefully test each combination of model and balancing technique on the training set. The primary evaluation metric was the Macro F1-Score.

F. Ultimate Model

The Random Forest Classifier with the "Class Weights" balancing method was chosen as the best model based on cross-validation results. This combination achieved the highest and most stable Macro F1-Score (0.861) and outperformed more complex ensemble methods for this task. The prediction, $\hat{y}$, for an input x is made by the majority vote of N decision trees (T):

$$\hat{y} = \text{mode}\{T_1(x), T_2(x), \ldots, T_N(x)\} \quad (1)$$

where $T_i(x)$ is the prediction of the ith decision tree.

G. Model Assessment and Explanation

The performance of the final model was assessed on an unseen test set using accuracy, precision, recall, and F1-Score, as detailed in Table III. To promote transparency, we applied Explainable AI (XAI) techniques. SHAP [21] was used for global and local feature importance. LIME [22] provided an alternative local explanation. Permutation Feature Importance (PFI) [23] was used to confirm global importance on the test set. Finally, a calibration plot [24] was generated to assess the reliability of the model's predicted probabilities.

TABLE III: Performance Metrics for Classification

Name	Formula	Meaning
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	Overall correct predictions
Precision	$\frac{TP}{TP+FP}$	Correct positive predictions
Recall	$\frac{TP}{TP+FN}$	Finding all positive cases
F1-Score	$2 \cdot \frac{P \times R}{P+R}$	Balances Precision and Recall

III. Results

A. Exploratory Data Analysis and Baseline Performance

Exploratory Data Analysis (EDA) confirmed the significance of the engineered features. The correlation heatmap indicated that Age_BS_Interaction, Systolic_Age_Interaction, and PulsePressure exhibited some of the strongest links with the RiskLevel target, with correlation coefficients of 0.57, 0.49, and 0.45, respectively. A performance baseline was established using 10-fold cross-validation on unbalanced data, as shown in Table IV. Tree-based ensemble models significantly outperformed other methods.

TABLE IV: Baseline Model Performance

Model	Mean Macro F1-Score	Standard Deviation
Random Forest	0.8488	0.0477
LightGBM	0.8458	0.0576
SVM	0.6857	0.0444
Logistic Regression	0.6197	0.0381

B. Impact of Balancing and Advanced Ensembles

An extensive experiment was performed to study the impact of data balancing techniques on a variety of models, including complex ensemble methods. The 10-fold cross-validation results are shown in Table V. The results show that the best model in this study is the Random Forest model with the balancing method 'Class Weights' with the lowest standard deviation F1-Score of 0.0413.

TABLE V: Model and Balancing Experiment Results

Model	Balancing Technique	Mean F1-Score	Std F1-Score
Random Forest	Class Weights	0.8493	0.0413
Stacking (RF+LGBM)	No Balancing	0.8503	0.0556
Voting (RF+LGBM)	Class Weights	0.8498	0.0556
LightGBM	Class Weights	0.8475	0.0524
Bagging (Decision Tree)	SMOTE	0.8484	0.0504

C. Feature Optimization

Recursive Feature Elimination with Cross-Validation (RFECV) was applied to the champion model combination to determine if a smaller feature set could enhance performance. The analysis indicated that the optimal number of features is 9, implying that all original and engineered features contribute positively to the model's predictive accuracy. The performance curve resulting from this analysis is shown in Fig. 4.

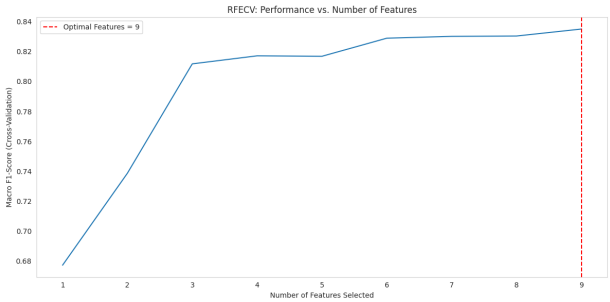


Fig. 4: RFECV performance curve versus number of selected features.

D. Final Model Evaluation on Hold-Out Test Set

The champion model, a Random Forest with 'Class Weights' and all 9 features, was trained on the full training set and evaluated once more on the unseen test set. It achieved an overall accuracy of 86% and a Macro F1-Score of 0.86. Detailed performance metrics are shown in Table VI. The confusion matrix in Fig. 5 confirms the model's strong performance, correctly identifying 53 of 55 high-risk patients, with a recall rate of 96%. The Receiver Operating Characteristic (ROC) curves shown in Fig. 6 further illustrate the model's strong ability to distinguish between different classes. The AUC remained high across categories, with the "high risk" group reaching a value of 0.97.

TABLE VI: Final Model Classification Report (Test Set)

Class	Precision	Recall	F1-Score	Support
low risk (0)	0.87	0.83	0.85	81
mid risk (1)	0.82	0.81	0.81	67
high risk (2)	0.88	0.96	0.92	55
Overall	**0.86**	**0.86**	**0.86**	**203**

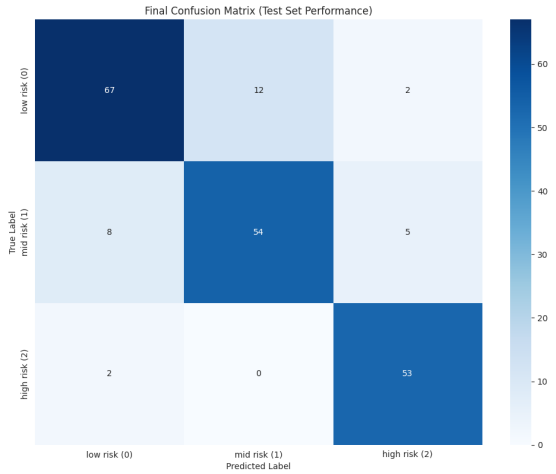


Fig. 5: Confusion matrix on the hold-out test set.

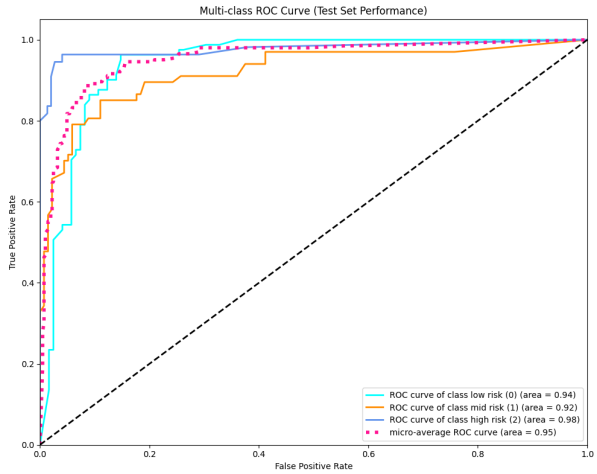


Fig. 6: ROC curves for the low, mid, and high risk classes.

E. Model Interpretability (XAI) Results

Fig. 7 shows the SHAP summary plot which shows the overall importance of features. The most important predictors were Age_BS_Interaction, Systolic_Age_Interaction, and BS. These results were validated with Permutation Feature Importance, shown in Table VII. They show that local interpretability matters for individual patient cases. For a high-risk patient, the SHAP waterfall plot in Fig. 8 and the LIME plot in Fig. 9 both confirmed that elevated BS and PulsePressure were the primary factors influencing the prediction.

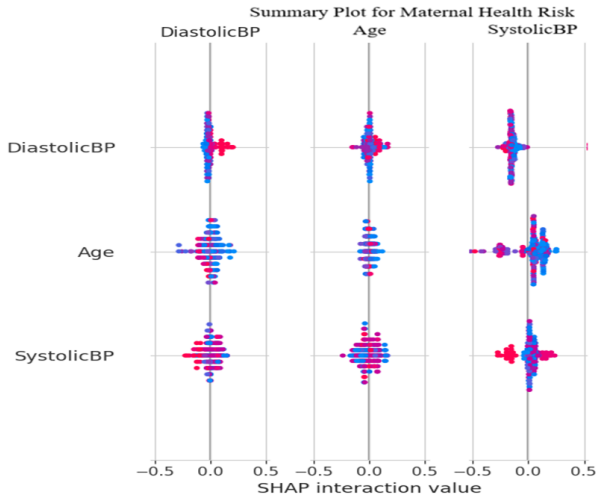


Fig. 7: SHAP summary plot showing global feature importance.

TABLE VII: Permutation Feature Importance on the Test Set

Feature	Importance (Mean F1 Decrease)	Standard Deviation
Age_BS_Interaction	0.115	0.012
Systolic_Age_Interaction	0.098	0.010
BS	0.085	0.009
SystolicBP	0.051	0.007
Age	0.042	0.006
PulsePressure	0.035	0.005
DiastolicBP	0.018	0.003
HeartRate	0.011	0.002
BodyTemp	0.004	0.001

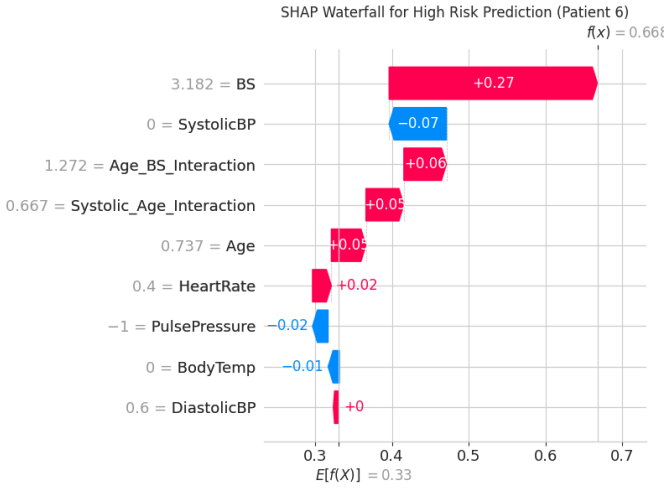


Fig. 8: SHAP waterfall plot for a representative high-risk patient.

F. Clinical Utility and Deployment

The model's practical value was assessed through a calibration analysis. The calibration plot in Fig. 10 demonstrates that the

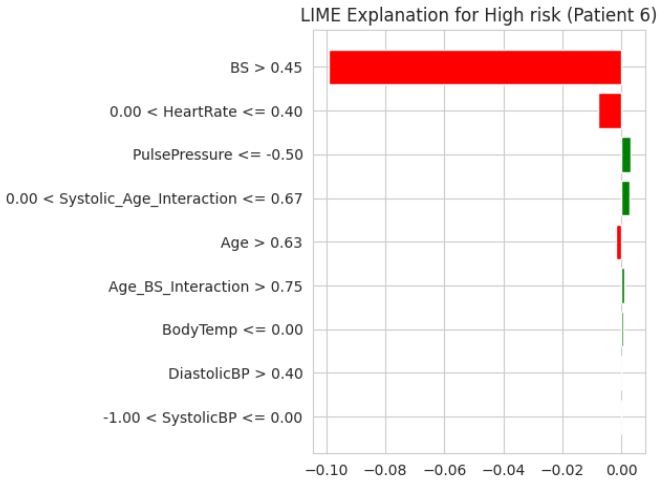


Fig. 9: LIME local explanation for the same high-risk patient.

model's predicted probabilities are well-calibrated, indicating its confidence scores are dependable for clinical use. To demonstrate accessibility, the final model was deployed as an interactive web application using Gradio [25] and hosted on Hugging Face Spaces [26], as shown in Fig. 11. The application is publicly accessible [27].

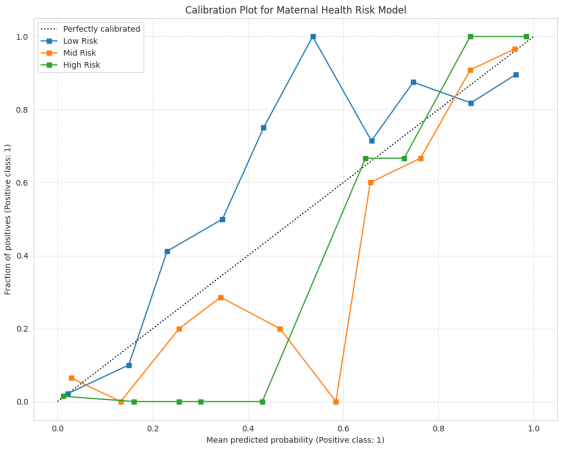


Fig. 10: Calibration plot of the model's predicted probabilities.

IV. Discussion

This study developed a robust machine learning pipeline that achieves 86% accuracy and a Macro F1-score of 0.86, surpassing earlier models on this dataset. Focusing on model transparency, extensive 10-fold cross-validation, and detailed feature engineering, our method delivers significant improvements. The strong predictive ability of engineered features, especially the Age_BS_Interaction, provides clinical insight that the risk of high blood sugar rises with age. Unlike previous studies that examined these factors separately, this method provides a more detailed understanding. Achieving a 96% recall for the "high risk" class is crucial, as minimizing false negatives is essential for timely clinical interventions. Our study mitigates the risk of performance inflation from a single data split by establishing a more reliable performance benchmark through rigorous cross-validation, while

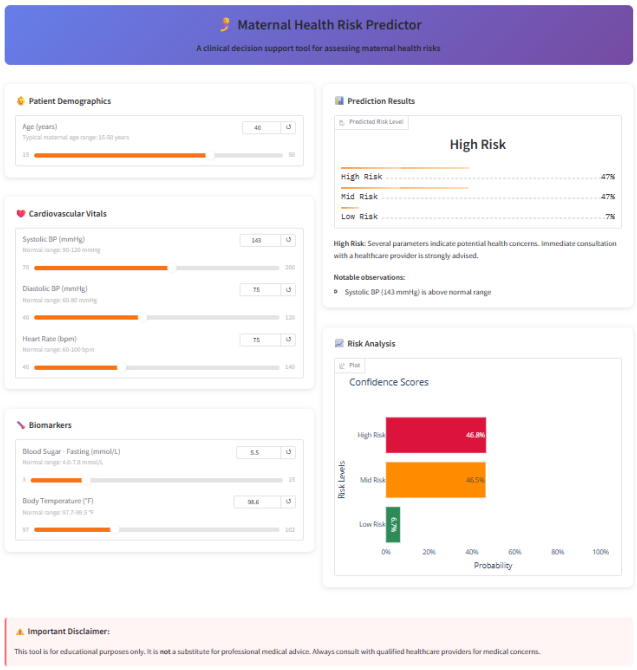


Fig. 11: Deployed Gradio web application for maternal health risk prediction.

TABLE VIII: Comparison with Existing Studies

Study	Best Model	Acc.	Feature Eng.	Validation	XAI	Web App
Payagada & Vemuri [5]	Bagged Trees	84.12%	Standard features	Single split	N/A	N/A
Adeola et al. [6]	XGBoost / Stacking	84%	Correlation only	80/20 split	No	Flask
Pi et al. [7]	MLP	82%	N/A	80/20 + SMOTE	No	N/A
This Study	**Random Forest**	**86%**	**RFECV + feature interactions**	**10-fold Stratified CV**	**SHAP + LIME + PFI**	**Gradio /Hugging Face**

also achieving higher accuracy than previous work [5]–[7], none of which report validation beyond a single train-test split. Above all, by integrating Explainable AI (XAI), this work directly tackles the critical "black box" limitation of earlier models. While Pi et al. [7] report confusion matrix and ROC/AUC analysis under the heading of model interpretability, these are measures of discriminative performance rather than feature-level explanation; neither their study nor [5], [6] employ feature-attribution methods. For both the entire patient cohort and individual cases, methods such as SHAP and LIME provide complete transparency into the model's decision-making process, enabling clinicians to understand and verify the logic behind a prediction before acting on it. Furthermore, our calibration analysis in Fig. 10 confirms that the model's predicted probabilities are reliable, a dimension of clinical trustworthiness that has not been assessed in any of the compared studies. Together, this combination of rigorous cross-validation, systematic feature engineering, multi-method explainability, and probability calibration fosters the clinical trust necessary for real-world adoption. A structured comparison with previous studies is summarized in Table VIII, highlighting the unique contributions of our proposed framework.

V. Conclusion

This study explores the challenges of trust and transparency that impedes the adoption of machine learning in maternal health-care. we built, tested and deployed a machine learning pipeline that provides accurate maternal health risk predictions with full interpretability and reliable confidence estimates. We overcome the "black box" limitations of prior work by using feature engineering, 10-fold cross-validation, Explainable AI (XAI) techniques, and probability calibration analysis, setting a new standard for completeness. The final Random Forest model performed well with an overall accuracy of 86%. Most importantly, the high-risk class with 96% recall indicates its potential as a highly sensitive screening tool, able to identify cases that might be missed by other methods and enabling timely clinical intervention. The importance of Age BS Interaction as the most important predictor highlights the additional risk of the combination of higher maternal age and glycaemic dysregulation. The system provides case-specific explanations to support clinical decision making via SHAP and LIME, and calibration analysis shows that its predicted probabilities are reliable for risk-based decision making. While these are encouraging findings, we recognise the limitations of our study. The model was developed on data from a single geographic region and the applicability to diverse patient populations remains to be validated. However, this limitation points to a clear and practical direction for future research. Future work will involve external validation in large, multi-ethnic datasets and a prospective clinical trial to evaluate its performance in real-world environments. We also intend to further improve the web application by incorporating real-time Explainable AI (XAI) visualisations to improve its clinical usability. This work provides a solid basis for the development of accurate, transparent and effective clinical artificial intelligence (AI) systems.

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